Notes: Braguinsky, Ohyama, Okazaki, and Syverson (AER, 2015)

Acquisitions, Productivity, and Profitability: Evidence from the Japanese Cotton Spinning Industry

Contents

1 Big picture

- **Question.** Do acquisitions improve productivity because more productive owners buy less productive plants, or is something subtler going on once we distinguish physical productivity from profitability and input utilization?
- **Setting.** The paper studies the universe of plants in the Japanese cotton spinning industry from 1896 to 1920, a period of rapid industrialization, large technological change, and intense acquisition activity.
- **Core challenge.** Standard merger studies usually observe revenue-based outcomes and capital or labor stocks, so they cannot cleanly separate physical productivity, prices, and utilization. This paper can, because it observes physical output, plant-level output prices, days of operation, service flows of labor and capital, ownership changes, and firm accounts.
- **Main empirical surprise.** Before acquisition, target plants were *not* on average less physically productive than the plants of future acquirers, conditional on operating. In fact, some acquired plants were more productive because they embodied newer capital.
- **Main profitability result.** Even though physical productivity was similar, target firms were much less profitable. The paper's central claim is that this reflects poorer **demand management**: higher inventories and accrued revenues, more idle time, and lower capacity utilization.
- **Main post-acquisition result.** After acquisition, target plants reduced unrealized output, raised utilization, increased quantity-based productivity, and became much more profitable. The gains are largest when the acquirer is one of the industry's serial acquirers.
- **Main mechanism.** The paper builds a simple model in which managers allocate scarce time between managing production and managing demand/sales. Better managers are better at demand management, so they can keep plants operating more intensively and also improve efficiency conditional on operation.
- **Big re-interpretation.** The right summary of acquisitions in this industry is not "higher productivity buys lower productivity." It is closer to **"higher profitability buys lower profitability"**, with better managers taking over better capital that had been used poorly.
- **Industry-level takeaway.** Between 1897 and 1914, industry TFP grew about 2.5% per year, while about 70% of capacity changed hands. The industry's growth involved not only entry and technology adoption but also reallocation of control toward firms that managed demand better.
- **Why this paper matters.** It is a benchmark paper on the link between acquisitions, management, and productivity. It shows that profitability gaps can matter more than pre-acquisition productivity gaps and that utilization is a central missing object in many merger studies.

2 Data and measurement (key objects)

2.1 Industry background and plant-level panel

- **Industry.** Japanese cotton spinning at the turn of the twentieth century, one of the classic cases of successful late industrialization.

- **Stages of development.** The paper describes three stages: an early stage with small and unproductive mills, a rapid 1890s growth stage after imported raw cotton and newer machinery were adopted, and then a twentieth-century consolidation stage with heavy acquisition activity.
- **Acquisition intensity.** There were 73 distinct acquisition deals involving 95 plant transfers between 1898 and 1920. Of the 78 plants operating in 1897, 49 were eventually acquired at least once, accounting for 63% of plants and 68% of capacity.
- **Unit of observation.** Plant-year.
- **Main operational data.** For each plant the authors observe the number of operating days, average daily spindles in operation, average daily male and female factory-floor employment, wages by gender, raw-cotton use, physical yarn output, yarn count, and average output price.
- **Main ownership data.** The plant's owning firm in each year is observed, so outcomes can be linked directly to acquisitions and ownership turnover.

2.2 Financial data and constructed variables

- **Firm accounts.** Semiannual company reports provide balance sheets, profit-and-loss statements, shareholder lists, and executive-board membership.
- **Productivity measure.** The benchmark is physical total factor productivity, denoted **TFPQ**, estimated from a production function using physical output and *service flows* of capital and labor rather than stocks.
- **Why service flows matter.** Because the data include the number of operating days, the authors can measure spindle-days and worker-days. This lets them distinguish productivity conditional on operation from input utilization.
- **Profitability measures.** Before acquisitions, the paper uses firm net earnings divided by paid-in shareholder capital. Around acquisitions, it constructs plant-level return on capital employed, denoted **ROCE**, from plant net operating surplus divided by capital employed allocated to the plant.
- **Price controls.** Output is cotton yarn of different counts, so price comparisons must adjust for quality. The paper therefore works with raw plant price and also with a *count-adjusted* price residual.
- **Capital vintage.** Equipment age is built from the vintage composition of installed spindles. Firm age is also used as an upper-bound proxy for plant age.
- **Demand-management variables.** Finished-goods inventories, accrued revenues on delivered output, unrealized output (the sum of the two), and spindle-utilization rates.

2.3 Unobservables, timing, and empirical uses

- **Productivity decomposition.** The production-function residual is split into persistent productivity and an innovation, following De Loecker's framework.

- **Timing.** Capital and labor service flows are chosen before the period's productivity innovation is fully realized; cotton used in production acts as the proxy variable because it moves with realized productivity.
- **Event-time windows.** The paper uses three indicators around each acquisition: a late pre-acquisition dummy (the 2 years immediately before the deal), an early post-acquisition dummy (the first 3 years after), and a late post-acquisition dummy (all later post-acquisition years).
- **Main empirical exercises.** The paper studies (i) pre-acquisition cross-sectional differences between future acquirers, future targets, and future exiters; (ii) within-target before-after changes; (iii) within-acquisition comparisons of acquired and incumbent plants; (iv) accounting and TFP decompositions; and (v) evidence on networks, engineers, and a simple mechanism model.

3 Models and empirical specifications (all equations + interpretation)

3.1 Production function and productivity

The benchmark production function is

$$y_{it} = \beta_k k_{it} + \beta_l l_{it} + \beta_i i_{it} + \beta_a a_{it} + \omega_{it} + \varepsilon_{it},$$

where y is logged physical output, k and l are logged service flows of capital and labor, i is the change in logged installed spindle capacity, and a is logged plant capital age.

Productivity evolves as

$$\omega_{it+1} = g(\omega_{it}, acq_{it}) + \xi_{it+1},$$

and the baseline specification is

$$g(\omega_{it}, acq_{it}) = \sum_{j=1}^{3} \gamma_j \omega_{it}^j + \theta_1 lb\_acq_{it} + \theta_2 ea\_acq_{it} + \theta_3 la\_acq_{it}.$$

Capital and labor coefficients are identified from the moment condition

$$\mathbb{E}[\xi_{it} \mid k_{it}, l_{it}] = 0.$$

The estimated production-function coefficients on labor and capital service flows are about 0.323 and 0.738, respectively.

- **Interpretation.** This is the paper's physical productivity measure *conditional on operation*. The unusual data allow the paper to build TFPQ using actual service flows instead of stocks.
- **Why De Loecker (2013) matters here.** Input use changes around acquisition, so a naive OLS residual could confound changes in productivity with endogenous changes in utilization or input choice.
- **Role of acquisition dummies.** They allow productivity dynamics to shift around the ownership change, so post-acquisition productivity gains need not be mechanically absorbed into input adjustment.

3.2 Within-acquired-plant event-study specification

The main within-target specification is

$$y_{it} = \alpha + \theta_1 lb\_acq_{it} + \theta_2 ea\_acq_{it} + \theta_3 la\_acq_{it} + m_A + \mu_t + \varepsilon_{it},$$

where y_{it} can be TFPQ, plant ROCE, or count-adjusted log price residuals.

- **Interpretation.** This asks whether the same acquired plant changes after acquisition relative to its own earlier performance.
- **Fixed effects.** m_A are acquisition fixed effects, which collapse to plant fixed effects for singly acquired plants but remain well defined for multiply acquired plants.
- **Time effects.** μ_t absorb industry-wide shifts.
- **Acquisition year.** The acquisition year itself is excluded because some deals happen mid-year and the authors do not want to mix pre- and post-takeover operations.

3.3 Within-acquisition difference-in-differences specification

The paper also compares acquired plants to the incumbent plants already owned by the acquirer:

$$\bar{y}_{it} = \alpha_0 + \beta_1 AA_{it} + \beta_2 Acquired_i + \beta_3 (Acquired_i \times AA_{it}) + m_{it} + \mu_t + \varepsilon_{it}.$$

Here AA_{it} equals one after the acquisition in acquisition case A, and the outcome for incumbent plants is the average outcome across all incumbent plants in that acquisition case.

- **Interpretation.** This is the paper's closest analogue to a merger-specific diff-in-diff. It asks whether acquired plants improve relative to the plants already operated by the acquirer.
- **Key coefficient.** β_3 is the post-acquisition relative change in the acquired plant.
- **Why useful.** This controls for acquisition-specific circumstances and for any time-varying shocks common to both target and acquirer plants.

3.4 Profitability decomposition

Plant ROCE is decomposed as

$$\frac{\pi_i}{C_i} = \frac{(1-\nu)Y_i}{C_i} - \frac{w_i L_i}{C_i} - \frac{R_i}{C_i},$$

where π_i is plant operating income, Y_i is output value, ν is the share of intermediate and nonlabor operating costs in output value, $w_i L_i$ is wage cost, R_i is capital cost, and C_i is capital employed.

The paper then decomposes net output value per unit of capital employed as

$$\log\left(\psi \frac{Y_i}{C_i}\right) = \log(\psi p_i) + \log\left(\frac{\exp(\widehat{Y}_i)}{C_i}\right) + TFPQ_i,$$

where $\psi = 1 - \nu$ is the price margin, p_i is plant price, $\widehat{Y}_i$ is predicted output from the production function, and $TFPQ_i$ is the residual.

- **Interpretation.** The first decomposition asks whether profitability differences come from output value, wage cost, or capital cost. The second asks whether output value per unit of capital comes from price margins, input intensity, or TFPQ.
- **Why this matters.** It opens the "black box" of post-acquisition profitability growth and isolates whether utilization or conditional efficiency is doing the work.
- **Central message.** Before acquisition, the main difference is not price and not much TFPQ; it is the amount of inputs actually mobilized per unit of capital employed.

3.5 Alternative productivity measures: TFPQ, TFPQU, and TFPR

The paper contrasts three TFP concepts:

- **TFPQ:** physical productivity using service flows of capital and labor, conditional on operating.
- **TFPQU:** physical productivity using capital and labor *stocks*, so differences in utilization appear as productivity differences.
- **TFPR:** revenue productivity, equal to TFPQU plus logged plant-specific output price.
- **Interpretation.** TFPQ is the paper's preferred concept because it strips out both price differences and utilization differences.
- **Why compare all three?** Most conventional producer microdata do not allow TFPQ. Comparing TFPQ to TFPQU and TFPR shows how much standard productivity measures exaggerate acquisition gains by loading utilization into "productivity."

3.6 Mechanism model: demand management and plant performance

The paper's simple model starts from plant output

$$y = g(m)x\omega,$$

where x is a composite input, ω is plant quality, and m is the manager's time devoted to production management. Time is split between utilization-enhancing effort u and conditional-efficiency effort v, with

$$g(m) = \sqrt{uv}, \qquad m = u + v \leq \gamma,$$

so γ is the manager's effective time endowment, interpreted as **demand-management ability**.

The input choice for producing output y is

$$x^* = \frac{y}{\sqrt{uv}\,\omega},$$

and if the plant sells $\gamma - m$ units at price p, revenues are

$$r = p(\gamma - m).$$

Therefore the owner's time-allocation problem is

$$\max_{u,v}\; (\gamma - u - v)\left(p - \frac{1}{\sqrt{uv}\,\omega}\right).$$

At the optimum, $u = v = m/2$, so the problem becomes

$$\max_{m} \; (\gamma - m)\left(p - \frac{2}{\omega m}\right),$$

with solution

$$m(\gamma,\omega) = \sqrt{\frac{2\gamma}{p\omega}}, \qquad \pi(\gamma,\omega) = \frac{(\sqrt{\gamma p \omega} - \sqrt{2})^2}{\omega}.$$

The model implies the following three propositions:

- **Proposition 1.** Both productivity and profitability of acquired plants rise after acquisition.
- **Proposition 2.** After acquisition, profits rise by more than TFPQ.
- **Proposition 3.** Pre-acquisition TFPQ of an acquiring plant can be lower than that of an acquired plant even though pre-acquisition profitability of the acquiring plant is always higher.

- **Interpretation.** Better demand management lets managers spend enough time on sales to keep output moving while also devoting more time to plant operation and operational efficiency.
- **Why this matches the data.** Targets can have newer capital and thus high TFPQ conditional on operation, but still be less profitable because they use that capital too little. A more able acquirer raises both utilization and TFPQ once the plant is taken over.

4 Identification and instruments (what makes the estimates credible)

4.1 What makes the empirical design stronger than standard merger evidence

- **Physical versus revenue productivity.** The paper observes physical output and plant prices separately, so it can distinguish true quantity productivity from price-based revenue productivity.
- **Utilization is directly observed.** Because the data include days of operation and daily spindles in operation, the paper does not have to guess at utilization from energy use or residual variation in revenue.
- **Dynamic productivity estimation.** Using De Loecker's method reduces transmission bias from plants adjusting inputs around acquisitions.
- **Mechanism variables are observed.** Inventories, accrued revenues, and plant-level financial accounts make it possible to see *why* profitability differs, not just that it differs.

4.2 What identifies the acquisition patterns in the paper

- **Before-after comparisons within acquired plants.** These eliminate time-invariant plant heterogeneity.
- **Within-acquisition comparisons to incumbent plants.** These remove many acquisition-specific confounds by comparing targets to the acquirer's own plants.
- **Pre-trend evidence.** The paper reports that there are no obvious pre-trends in acquired plants' relative performance, while some performance measures show discrete changes around acquisition.

- **No strong causal claim from exogenous assignment.** The paper is explicit that acquisitions are not randomly assigned. The credibility comes from rich data, timing, fixed effects, and consistent patterns across multiple specifications and decompositions rather than from an external instrument.

4.3 What the decomposition evidence adds

- **Prices are mostly ruled out.** Count-adjusted prices do not explain the profitability gap before acquisition or its improvement after acquisition.
- **Market power is not the story.** The evidence is inconsistent with acquisition gains being driven mainly by higher output prices.
- **Utilization is central.** The paper shows that much of the apparent revenue-productivity improvement in conventional measures would actually be utilization improvement.
- **Demand-management proxies line up.** Networks with major traders and the presence of formally educated engineers are both strongly associated with lower unrealized output, higher utilization, and higher profitability.

4.4 Relation to the literature

- **Against the simple efficiency-merger view.** The paper rejects the crude idea that acquirers were simply more physically productive ex ante.
- **Relative to assortative matching stories.** The findings fit better with "better firms buy less profitable firms" and then raise profitability by using assets more effectively.
- **Connection to management literature.** The paper provides a concrete channel through which management quality affects plant outcomes: demand management changes both utilization and conditional productivity.

5 Tables and figures

5.1 Figure 1: Domestic output, import, and export of cotton yarn, 1887–1914

Read:

- The industry moves from import dependence to domestic dominance and then export strength.
- The 1890s are a period of explosive expansion, helped by imported longer-stapled cotton and newer machinery.
- The figure sets up the paper's acquisition story: once imports are displaced and the market becomes tighter, overproduction and consolidation become central.
- This is the macro backdrop for why acquisitions became such an important reallocation mechanism after 1898.

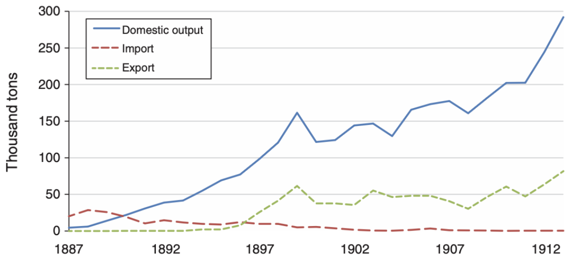

Figure 1. Domestic Output, Import, and Export of Cotton Yarn, 1887–1914

Source: Nihon Choki Tokei Soran, authors' estimates.

Table 1—Future Acquiring, Acquired, and Exiting Plants in 1896–1897

		Acquired plants		
	Acquiring plants	First cohort	Second cohort	Exiting plants
TFPQ	0.066	0.034	0.156	−0.211
	(0.156)	(0.225)	(0.229)	(0.552)
Profit per paid-in share	0.274	0.185	0.159	0.159
	(0.205)	(0.074)	(0.149)	(0.101)
Price (yen/400 lb.)	94.7	92.4	92.8	91.7
	(6.5)	(3.8)	(7.4)	(7.0)
Logged price residual	−0.041	0.014	0.012	0.021
	(0.151)	(0.040)	(0.041)	(0.062)
Main count of yarn produced	21.5	17.5	17.2	13.9
	(11.5)	(2.6)	(4.7)	(5.6)
Days in operation	323.7	315.9	300.6	278.6
	(29.8)	(29.5)	(55.6)	(56.8)
Equipment age	5.28	5.88	2.79	11.77
	(3.49)	(2.76)	(1.00)	(6.69)
Firm age	9.13	11.06	3.31	12.54
	(5.08)	(3.81)	(2.05)	(7.86)
Observations	32	33	32	24

Notes: This table reports plant-level means and standard deviations (in parentheses) of data for various subsamples of plants. "Acquiring plants" are those owned by future acquiring firms, "acquired plants" are owned by firms that will be acquired in the future, and "exiting plants" are owned by firms that will exit in the future (not through acquisition) and be scrapped. "First cohort" are plants of firms that started operating before 1892, "second cohort" is plants of firms that started operating in or after 1892. 1896 and 1897 observations for second-cohort plants that began operations in those years are excluded. TFPQ (quantity-based total factor productivity) is estimated using De Loecker's (2013) method described in the main text. Profit per paid-in share is net revenue from company reports divided by shareholders' paid-in capital. There are only six observations on net revenue available for exiting plants in these years. The log price residuals are from a regression of log plant-level price on count dummies and year dummies, as described in the main text. Equipment and firm age are measured in years.

Source: Authors' calculations.

5.2 Table 1: Future acquiring, acquired, and exiting plants in 1896–1897

Read:

- Future acquirers are the most profitable plants before any acquisitions happen: profit per paid-in share is 0.274 versus 0.185 for first-cohort targets and 0.159 for second-cohort targets.
- But acquirers are *not* the most physically productive. TFPQ is 0.066 for future acquirers, 0.034 for first-cohort targets, and 0.156 for second-cohort targets.
- Prices are not the source of the profitability gap. Count-adjusted log price residuals are close to zero for all groups.
- The real differences are utilization and capital vintage: future acquirers operate more days per year (323.7 versus 300.6 for second-cohort targets), while second-cohort targets have younger capital (equipment age 2.79 versus 5.28 for acquirers).

5.3 Table 2: Within-acquired-plants comparisons of productivity, profitability, and prices

Read:

- For all acquisitions, acquired plants' TFPQ rises by about 4.5% in the first three years and by about 13.4% in the longer run.
- Profitability improves even faster: plant ROCE rises by about 6 percentage points early and 9 percentage points in the longer run.
- Count-adjusted prices barely move, so post-acquisition profitability growth is not driven by output price increases.
- For serial acquirers, the gains are larger: long-run TFPQ rises about 17.2%, and plant ROCE rises about 14 percentage points.

Table 2—Within-Acquired-Plants Comparisons of Productivity, Profitability, and Prices

	All acquisitions			By serial acquirer		
	TFPQ	Plant ROCE	log price residuals	TFPQ	Plant ROCE	log price residuals
Late pre-acquisition dummy	−0.003	0.020	0.011	−0.016	0.025	0.018
	(0.019)	(0.013)	(0.013)	(0.030)	(0.016)	(0.030)
Early post-acquisition dummy	0.045*	0.060***	0.036	0.053	0.106***	0.065
	(0.026)	(0.022)	(0.027)	(0.046)	(0.023)	(0.063)
Late post-acquisition dummy	0.126***	0.089***	0.044	0.159**	0.140***	0.089
	(0.033)	(0.025)	(0.034)	(0.062)	(0.032)	(0.068)
Constant	0.603***	0.102***	0.029***	0.356***	0.079**	0.041***
	(0.032)	(0.013)	(0.010)	(0.025)	−0.031	(0.008)
Acquisition fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,078	891	1,118	512	472	521
Adjusted R^2	0.734	0.639	0.097	0.695	0.625	0.082

Notes: The omitted category is period three years or more prior to acquisition. Serial acquirers are Kanegafuchi, Mie, Osaka, Settsu, and Amagasaki Boseki. The omitted category includes the period three years or more prior to acquisition. Robust standard errors clustered at the acquisition level in parentheses.
***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Source: Authors' calculations.

Table 3—Within-Acquisition Comparisons of Productivity and Profitability: Acquired and Incumbent Plants

	All acquisitions			By serial acquirer		
	TFPQ	Plant ROCE	log price residuals	TFPQ	Plant ROCE	log price residuals
After acquisition	−0.055***	−0.004	−0.031**	−0.048***	−0.012	−0.026*
	(0.013)	(0.012)	(0.013)	(0.008)	(0.016)	(0.015)
Acquired plant	−0.025	−0.030***	−0.019	−0.032*	−0.032**	−0.015
	(0.021)	(0.011)	(0.014)	(0.017)	(0.013)	(0.015)
After acquisition × acquired plant	0.091***	0.040***	0.024	0.113***	0.058***	0.033
	(0.023)	(0.014)	(0.017)	(0.028)	(0.017)	(0.025)
Constant	0.480***	0.145***	0.038***	0.410***	0.069***	−0.007
	(0.034)	(0.018)	(0.008)	(0.008)	(0.013)	(0.008)
Acquisition fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,487	1,392	1,528	1,067	994	1,091
Adjusted R^2	0.347	0.433	0.108	0.489	0.455	0.164

Source: Authors' calculations. See Notes for Table 2.

5.4 Table 3: Within-acquisition comparisons of acquired and incumbent plants

Read:

- Relative to incumbent plants of the acquirer, acquired plants' TFPQ rises by about 9.1% in the full sample and 11.3% for serial acquirers.
- Relative plant ROCE rises by about 4 percentage points in the full sample and about 5.8 percentage points for serial acquirers.
- Before acquisition, acquired plants are not systematically less productive than incumbent plants, but they are significantly less profitable.
- Again, prices show little change, reinforcing that the post-acquisition gains are about productivity and utilization rather than market power.

5.5 Table 4: Decomposition of returns on capital employed

Read:

- Before acquisition, incumbent plants' ROCE exceeds that of acquired plants by 0.051, or about 95%.
- Most of this gap comes from net output value relative to capital employed: that term is higher by 0.065 for incumbents, while wage costs are actually higher for incumbents and therefore work against the ROCE gap.
- After acquisition, ROCE rises by 0.063 in the early post period and by 0.100 in the late post period.
- The profit gains are driven mainly by higher net output value per unit of capital employed, not by lower wages or large drops in capital cost.

5.6 Table 5: Decomposition of net output values

Read:

TABLE 4—DECOMPOSITION OF PLANTS' RETURNS ON CAPITAL: INCUMBENT AND ACQUIRED PLANTS, AND ACQUIRED PLANTS PRE- AND POST-ACQUISITION

Pre-acquisition means	Acquired plants (A)	Incumbent plants (B)	Difference (B) − (A)	Percentage difference
ROCE	0.053	0.104	0.051	95.3***
Of which:				
net output value/capital employed	0.193	0.257	0.065	33.5***
Minus:				
wage cost/capital employed	0.077	0.094	0.018	22.9***
capital cost/capital employed	0.062	0.059	−0.004	−6.2***
Observations	133	269		
Pre- and early post-acquisition means	**Pre-acquisition (A)**	**Early post-acquisition (B)**	**Difference (B) − (A)**	**Percentage difference**
ROCE	0.062	0.126	0.063	101.7***
Of which:				
net output value/capital employed	0.202	0.286	0.084	41.6***
Minus:				
wage cost/capital employed	0.078	0.103	0.025	32.2***
capital cost/capital employed	0.062	0.058	-0.004	−7.0**
Observations	163	159		
Pre- and late post-acquisition means	**Pre-acquisition (A)**	**Late post-acquisition (B)**	**Difference (B) − (A)**	**Percentage difference**
ROCE	0.062	0.163	0.100	161.1***
Of which:				
net output value/capital employed	0.202	0.317	0.114	56.6***
Minus:				
wage cost/capital employed	0.078	0.103	0.025	31.7***
capital cost/capital employed	0.062	0.051	−0.011	−17.3***
Observations	163	280		

Notes: The "pre-acquisition" time period includes observations on up to four years prior to acquisition. "Early post-acquisition" period includes three years immediately following acquisitions. "Late post-acquisition" period includes years starting from year 4 after acquisitions. Nominal variables are deflated by the annual consumer price index. Details of variable construction are explained in online Appendix E.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

Source: Authors' calculations.

TABLE 5—DECOMPOSITION OF PLANTS' NET OUTPUT VALUES: INCUMBENT AND ACQUIRED PLANTS AND ACQUIRED PLANTS PRE- AND POST-ACQUISITION

Pre-acquisition means of logs	Acquired plants (A)	Incumbent plants (B)	Difference (B) − (A)	Percentage difference
ln(net output value/capital employed)	−1.791	−1.436	0.355	42.6***
Of which:				
ln(price margin)	−1.407	−1.377	0.030	3.1
TFPQ	0.500	0.568	0.069	7.1***
ln(total input/capital employed)	−0.883	−0.627	0.256	29.2***
Observations	129	262		
Pre- and early post-acquisition means of logs	**Pre-acquisition (A)**	**Early post-acquisition (B)**	**Difference (B) − (A)**	**Percentage difference**
ln(net output value/capital employed)	−1.735	−1.392	0.343	40.9***
Of which:				
ln(price margin)	−1.438	−1.367	0.071	7.4**
TFPQ	0.499	0.568	0.069	7.2***
ln(total input/capital employed)	−0.795	−0.593	0.202	22.4**
Observations	157	157		
Pre- and late post-acquisition means of logs	**Pre-acquisition (A)**	**Late post-acquisition (B)**	**Difference (B) − (A)**	**Percentage difference**
ln(net output value/capital employed)	−1.735	−1.275	0.460	58.4***
Of which:				
ln(price margin)	−1.438	−1.316	0.122	13.0***
TFPQ	0.499	0.685	0.187	20.5***
ln(total input/capital employed)	−0.795	−0.644	0.151	16.3***
Observations	157	278		

Notes: The "pre-acquisition" time period includes observations on up to four years prior to acquisition. "Early post-acquisition" period includes three years immediately following acquisitions. "Late post-acquisition" period includes years starting from year 4 after acquisitions. Nominal variables are deflated by the annual consumer price index. Details of variable construction are explained in online Appendix E.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

Source: Authors' calculations.

- Before acquisition, the log gap in net output value relative to capital employed is 0.355, or about 42.6%.
- Only a small part of that comes from price margins (0.030) and TFPQ (0.069). The biggest part is total input relative to capital employed (0.256), meaning better firms mobilize much more input flow per unit of capital.
- In the early post-acquisition period, profitability growth still comes mainly from higher input intensity (0.202), with TFPQ contributing 0.069.
- In the late post-acquisition period, TFPQ becomes more important (0.187), but input intensity still explains a substantial part (0.151).

5.7 Table 6: TFP measures around acquisition events

Read:

- Using conventional revenue productivity (TFPR), acquisition gains look much larger than under the benchmark TFPQ measure.
- Within acquired plants, TFPR rises by about 16.8% early and 29.0% late, compared with TFPQ gains of 4.5% and 12.6%.
- The intermediate measure TFPQU, which uses stocks rather than service flows, tracks TFPR closely. This shows that most of the gap between TFPR and TFPQ comes from utilization, not from prices.
- The table demonstrates why standard producer microdata would overstate the productivity effects of acquisitions in this setting.

TABLE 6—TOTAL FACTOR PRODUCTIVITY CHANGES AROUND ACQUISITION EVENTS

	Within-acquired plants estimations				Difference-in-differences estimations		
Dependent variable	TFPR	TFPQU	TFPQ	Dependent variable	TFPR	TFPQU	TFPQ
Late pre-acquisition dummy	0.020	−0.027	−0.003	After acquisition	−0.083**	−0.042*	−0.055***
	(0.051)	(0.044)	(0.019)		(0.035)	(0.024)	(0.013)
Early post-acquisition dummy	0.168***	0.104***	0.045*	Acquired plant	−0.075**	−0.093***	−0.025
	(0.058)	(0.038)	(0.026)		(0.035)	(0.029)	(0.021)
Late post-acquisition dummy	0.290***	0.211***	0.126***	(After acquisition) × (acquired plant)	0.148***	0.139***	0.091***
	(0.080)	(0.050)	(0.033)		(0.042)	(0.033)	(0.023)
Constant	0.750***	0.304***	0.603***	Constant	1.197***	0.393***	0.480***
	(0.065)	(0.053)	(0.032)		(0.083)	(0.042)	(0.034)
Acquisition fixed effects	Yes	Yes	Yes	Acquisition fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Year fixed effects	Yes	Yes	Yes
Observations	1,047	1,077	1,078	Observations	1,430	1,486	1,487
Adjusted R^2	0.824	0.478	0.734	R^2	0.636	0.318	0.347

Notes: TFPQ is our benchmark TFP measure that uses capital and labor services flows as inputs. TFPQU is unconditional TFPQ, using instead plants' total capacity and labor input. TFPR equals TFPQU plus logged plant-specific output price. The omitted category includes period 3 years or more prior to acquisition. The omitted category includes period 3 years or more prior to acquisition. Robust standard errors clustered at the acquisition event level in parentheses.

***Significant at the 1 percent level.
**Significant at the 5 percent level.
*Significant at the 10 percent level.

Source: Authors' calculations.

TABLE 7—INVENTORY AND ACCRUED PAYMENTS TO OUTPUT VALUE RATIOS: INCUMBENT AND ACQUIRED PLANTS AND ACQUIRED PLANTS PRE- AND POST-ACQUISITION

Means	Acquired plants (A)	Incumbent plants (B)	Difference (B − A)	Percentage difference
Inventory/produced output (C)	0.046	0.018	−0.028	−61.0***
Accrued revenues/produced output (D)	0.031	0.015	−0.016	−50.6***
Unrealized/produced output (C) + (D)	0.078	0.033	−0.045	−57.4***
Observations	113	195		
	Pre-acquisition (A)	Early post-acquisition (B)	Difference (B − A)	Percentage difference
Inventory/produced output (C)	0.048	0.013	−0.034	−72.0***
Accrued revenues/produced output (D)	0.029	0.020	−0.010	−32.4**
Unrealized/produced output (C) + (D)	0.078	0.032	−0.046	−59.4***
Observations	139	100		
	Pre-acquisition (A)	Late post-acquisition (B)	Difference (B − A)	Percentage difference
Inventory/produced output (C)	0.048	0.009	−0.039	−81.5***
Accrued revenues/produced output (D)	0.029	0.015	−0.014	−48.1***
Unrealized/produced output (C) + (D)	0.078	0.023	−0.055	−70.6***
Observations	139	124		

Notes: The "pre-acquisition" time period includes observations on up to 4 years prior to acquisition. "Early post-acquisition" period includes 3 years immediately following acquisitions. "Late post acquisition" period includes years starting from year 4 after acquisitions.

***Significant at the 1 percent level.
**Significant at the 5 percent level.

Source: Authors' calculations.

5.8 Table 7: Inventory and accrued-payment ratios

Read:

- Before acquisition, acquired plants have much more unrealized output: the unrealized-output-to-produced-output ratio is 0.078 for targets versus 0.033 for incumbents.
- In the first three years after acquisition, the unrealized-output ratio falls by about 59.4%.
- In the later post-acquisition period, it falls by about 70.6% relative to the pre-acquisition level.
- This is some of the strongest direct evidence for the paper's demand-management story.

5.9 Table 8: In-network versus out-of-network firms

Read:

- Firms connected to major traders have higher TFPQ (0.488 versus 0.433), much higher TFPQU (0.241 versus 0.117), and much higher ROCE (0.059 versus 0.023).
- Their unrealized-output ratio is lower by 0.043, and spindle-utilization rates are higher by 0.043.
- These firms also have slightly higher count-adjusted prices, but the main difference is not price; it is lower inventory/arreares and higher utilization.
- The future-acquirer probability is dramatically higher for in-network firms, which ties trader relationships directly to the ownership-reallocation process.

5.10 Table 9: Firms with and without formally educated engineers

Read:

- Firms with formally educated engineers have higher TFPQ (0.517 versus 0.435), higher TFPQU (0.286 versus 0.131), and higher ROCE (0.072 versus 0.024).

Table 8—Plant and Firm Performance Metrics, 1898–1902: In-Network and Out-of Network Firms

Outcome	Out-of-network (A)	In-network (B)	Difference (B − A)
TFPQ	0.433	0.488	0.055***
TFPQU	0.117	0.241	0.123***
ROCE	0.023	0.059	0.037***
Unrealized output ratios	0.127	0.084	−0.043***
Spindle utilization rates	0.739	0.781	0.043**
Logged price residuals	−0.025	0.018	0.044***
Observations	127	170	

***Significant at the 1 percent level.
**Significant at the 5 percent level.
Source: Authors' calculations.

Table 9—Plant and Firm Performance Metrics in 1898–1902 by Firms with and without Educated Engineers

Outcome	No formally educated engineer (A)	Formally educated engineer (B)	Difference (B − A)
TFPQ	0.435	0.517	0.082***
TFPQU	0.131	0.286	0.156***
ROCE	0.024	0.072	0.047***
Unrealized output ratios	0.119	0.078	−0.042***
Spindle utilization rates	0.746	0.792	0.046***
Logged price residuals	−0.014	0.021	0.035**
Observations	188	109	

***Significant at the 1 percent level.
**Significant at the 5 percent level.
Source: Authors' calculations.

- They also have lower unrealized-output ratios and higher spindle-utilization rates.
- The educated-engineer pattern looks very similar to the network pattern, which suggests that technical competence and demand management were complementary advantages.
- Importantly, many of the firms with educated engineers were also in-network firms, including all five serial acquirers.

6 Short summary

- **Conclusion.** Acquisitions in this industry improved both productivity and profitability, but not because acquirers were simply buying low-productivity plants.
- **Main insight.** Targets often had good capital and decent physical productivity conditional on operation, yet poor profitability because they managed demand badly and therefore underutilized capacity.
- **Main empirical lesson.** Once more profitable firms took control, targets reduced inventories and accrued revenues, operated more intensively, and became both more productive and more profitable.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Ownership change can raise productivity and profitability through better management of existing assets, even when targets are not initially less physically productive than acquirers.
- **Methodological contribution.** The paper combines unusually rich microdata with a productivity estimator that separates conditional efficiency from utilization and prices.
- **Mechanism contribution.** The paper identifies a new channel in the acquisitions literature: superior **demand management** increases utilization, lowers unrealized output, and eventually lifts productivity conditional on operation.
- **Quantitative contribution.** Post-acquisition gains are economically large, especially for acquisitions by the industry's serial acquirers.

7.2 Economic intuition

- A plant can embody good capital and still perform badly if management cannot keep sales flowing, collect payments, and operate the factory steadily.
- Better firms in this industry were not necessarily transforming inputs into output more efficiently at every moment they operated; they were much better at making sure the plant *did operate* and that output got sold and paid for.
- Once such firms acquired target plants, they spread that ability across the acquired capital. This immediately improved utilization and profitability and then, over time, also improved TFPQ.
- That is why profitability can be a better predictor of acquisition than measured productivity in settings where utilization and demand management are central.

7.3 Takeaway

This paper is best read as a management-and-reallocation paper rather than a simple merger-efficiency paper. Its most important lesson is that plant performance depends not only on the technology embodied in capital, but also on how well managers keep that capital in use and connected to demand. When more capable firms take over underutilized but potentially strong assets, the result can be large gains in profitability first and productivity next. That is the paper's core contribution, and it is why this study remains a model case for thinking carefully about productivity, profitability, and ownership together.